[INCH-POUND]
A-A-59477
30 September 1999
SUPERSEDING
MIL-G-80048C
22 July 1983

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINES, CENTERLESS, EXTERNAL, CYLINDRICAL

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-G-80048C for all federal agencies.

1. SCOPE. This commercial item description (CID) covers external cylinder centerless grinding machines intended for grinding external surfaces of plain, tapered, or contoured cylinders, where centers or chucks can not be used or are impractical to hold the workpiece.

2. CLASSIFICATION. The grinding machines shall be of the following sizes. Size is determined by the diameter of the workpiece and width of the grinding wheel as shown in table I. The size to be furnished shall be as specified (see 7.3(b)).

3. SALIENT CHARACTERISTICS

3.1 General Requirements. The grinding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The grinding machine shall meet the size and capacity requirements shown in table I. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

3.1 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Frame. The frame shall provide a means to permit leveling and mounting the grinding machine to the floor or foundation.

3.2.2 Bed. The bed shall support the other elements of the machine, have sufficient mass to dampen adverse vibration, and provide a means for leveling the bed surface. The bed surface shall not deviate from level by more than 0.004 inch per 40 inches measured. When applicable, access doors or removable plates shall be provided for any components housed within the bed.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Machine sizes and capacities.

Size	Workpiece diameter, inches	Grinding wheel	
		Width, inches	Horsepower
01	0.05 – 1.5	3	7.5
02		4	7.5
03		6	7.5
04		8	10
05	0.7 – 2.0	4	10
06		5	10
07		6	10
08		8	10
09		10	15
10	0.125 – 4.0	6	20
11		8	20
12		10	20
13		12	30
14		16	30
15	0.2 – 5.0	8	30
16		10	30
17		12	30
18		14	50
19		20	50
20	0.5 – 6.0	12	30
21		20	30
22		24	50
23		28	50
24	0.5 – 8.0	12	40
25		14	60
26		16	60
27		20	74
28		22	74
29		26	74

NOTE: The values shown are minimum requirements.

3.2.3 Coolant system. The coolant system shall incorporate an adjustable nozzle and shall provide adequate capacity to maintain a steady flow of clean coolant over the work area. Coolant flow rate shall be sufficient to prevent workpiece discoloration or loss of temper during the grinding operation. When specified (see 7.3(c)), optional filtering and temperature control equipment shall be provided.

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3.2.4 Grinding wheel head assembly. The grinding wheel head assembly shall include a spindle, wheel dresser, and a grinding wheel. The assembly shall be equipped with wheel guards and access covers that allow easy removal and/or replacement of the grinding wheel. Grinding wheel spindles shall be supported by bearings that are either sealed for life or automatically lubricated. The spindle motor shall meet or exceed the horsepower requirements shown in table I and shall be capable of continuous operation at full capacity (i.e., largest workpiece diameter, widest grinding wheel width, and highest grinding speed) while operating at not more than 85 percent of the rated horsepower. Unless otherwise specified (see 7.3(d)), manual infeed shall be provided when an adjustable wheel head assembly is incorporated in the machine design. Manual adjustment shall be controlled by a graduated handwheel with a wheel feed index. Each graduation shall indicate not more than 0.001 inch on workpiece diameter.

3.2.5 Regulating wheel head assembly. The regulating wheel head assembly shall include a spindle, wheel dresser, and regulating wheel. The assembly shall be equipped with wheel guard and access covers that allow easy removal and/or replacement of the regulating wheel. Regulating wheel spindles shall be supported by bearings that are either sealed for life or automatically lubricated. Unless otherwise specified (see 7.3(e)), manual infeed shall be provided when an adjustable regulating wheel head is incorporated. Manual adjustment shall be controlled by a handwheel with a wheel feed index. Each graduation shall indicate not more than 0.001 inch on workpiece diameter. The tilting (vertical axis) and swiveling (horizontal axis) actions shall be adjustable with positive clamping. A scale, graduated in degrees, shall be provided for measuring tilt. A means for the fine correction of taper shall be provided. The regulating wheel width shall be equal to or greater than the width of the grinding wheel.

3.2.6 Dressing/truing assembly. A dressing/truing assembly shall be provided for each wheel (grinding, regulating) and shall meet the requirements shown in table II. A micrometer-type indicator or dial graduated in no greater than 0.001-inch increments shall be provided for designating movement. Unless otherwise specified (see 7.3(f)), the range of dressing depth shall be not less than 0 to 2.5 inches for regulating wheels and 0 to 3.5 inches for grinding wheels. A powered feed mechanism shall be provided for traversing of the dresser across the wheel. Unless otherwise specified (see 7.3(g)), the dresser shall be capable of contour dressing.

3.2.7 Workrest blade holder assembly. The grinding machine shall be provided with a workrest blade holder assembly. At least one set of blades appropriate for the performance test shall be provided. Additional workrest blades shall be provided as specified (see 7.3(h)).

3.2.8 Slide assembly. Slide assemblies for the regulating wheel and/or workrest blade holder must meet the requirements shown in table II.

3.2.9 Regulating wheel tilt and swivel. Unless otherwise specified (see 7.3(i)), regulating wheels with a width of 12 inches or less shall tilt a minimum of +4/–1 degrees, and shall swivel at least 4 degrees total. Swivel shall be not more than half in the negative direction (i.e., +4/–0 degrees and +2/–2 degrees are acceptable). Unless otherwise specified (see 7.3(j)), regulating wheels with a width over 12 inches shall tilt a minimum of +1/–1 degrees and shall swivel a minimum of 2 degrees total. Swivel shall be not more than 1/2 degree in the negative direction.

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TABLE II. Geometric alignment requirements.

Component	Test	Measuring distance, inches	Maximum deviation, inch
Bed surface	Level	40	0.004
Grinding wheel head assembly spindle	Run-out, wheel mount	Full rotation	0.0002
	Axial slip	Full rotation	0.00032
	Run-out, bearings	Full rotation	0.0002
Regulating wheel head assembly spindle	Run-out, wheel mount	Full rotation	0.0002
	Axial slip	Full rotation	0.0004
	Run-out, bearings	Full rotation	0.0002
Grinding wheel dressing assembly	Straightness	12	0.00012
	Parallelism	12	0.002
Regulating wheel dressing assembly Slide assembly	Straightness	12	0.00012
	Parallelism Straightness ways	12	0.0012
		40	0.0008
		12	0.00024
	Straightness motion	40	0.0004
	Position repeatability	[1]Maximum work diameter	0.00008

[1] Travel from the maximum workpiece to wheel disengagement to the point of workpiece contact at maximum workpiece diameter.

TABLE III. Performance requirements.

Machine size	Test rod diameter, inches	Test rod length	Maximum diameter deviation, inch	Maximum circularity deviation, inch
00 - 04	0.5	90% of grinding wheel width	0.00008	0.00008
05 - 12	1.0		0.0001	
13 - 24	2.0		0.00015	
25 - 30	4.0		0.0002	

3.2.10 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Machinery, Industrial, Electrical Standard For". The machine shall operate from a dual 230/460 volt, 3-phase, 60-hertz circuit and shall be initially wired for the voltage specified (see 7.3(k)). The control system shall operate on a 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.11 Electric motors. The grinding machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.3(l)), the motors shall have dripproof enclosures.

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3.2.12 Optional equipment. Optional equipment shall be furnished as specified (see 7.3(m)).

3.3 Performance. The workpiece shall have a surface finish equal to or better than 0.00003 inch when tested in accordance with ANSI B46.1, "Surface Texture (Surface Roughness, Waviness, Lay)". The machine shall meet the requirements specified in tables II and III when tested in accordance with sections 5.3 through 5.3.4 herein.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.3(n)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9, "Machines, Grinding, Safety Requirements for the Construction, Care, And Use Of". The manufacturer shall ensure the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "General Industry, Safety and Health Standards". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular system required shall be as specified (see 7.3(o)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.3(p)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.3(q)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer’s name
c. Manufacturer’s model designation
d. Power input (volts, total amperes, phase, frequency)
e. Short-circuit/over-current protection rating
f. Contract number or order number
g. National stock number
h. Date of manufacturer

3.8 Lubrication plate. When specified (see 7.3(r)), a lubrication plate shall be attached to each machine. The plate shall include the following information:

a. Points of lubrication
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The grinding machine shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Testing. Unless otherwise specified (see 7.3(s)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.2.1 Test conditions. Unless otherwise specified (see 7.3(t)), the grinding machine shall be tested in accordance with the environmental conditions specified by the manufacturer.

5.2.2 Operational test. The external cylinder centerless grinding machines shall be operated in accordance with the manufacturer's standard operating procedures. The grinders shall be operated at no load for not less than 3 hours. Proper operation of all controls and features shall be verified during the trial period.

5.2.3 Performance test. The machine shall be tested by grinding three rods of C-1045 steel using the infeed (plunge) method to remove 0.01 inch of material. The size test rod shall be selected using table III guidelines in accordance with the size of the machine tested. Machine warm-up, setup, and grinding rate shall follow manufacturer's recommended procedure. In the event the machine supplied is capable of only thrufeed operation, the test rods shall be ground using the thrufeed method.

5.2.4 Geometric alignment test. The geometric alignment testing shall be conducted in accordance with tests G1, G2, G4, and G5 of ISO 3875, "Acceptance Conditions for External Cylindrical Centreless Grinding Machines - Testing of the Accuracy", and G2 and G13 of ISO 2433, "Acceptance Conditions for External Cylinder Grinding Machines with a Movable Table - Testing of Accuracy". When contour dressing is specified, deviation from the template shall be equal to or better than the specification for straightness shown in table II.

5.3 Acceptance criteria. Unless otherwise specified (see 7.3(u)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The geometric alignment test results shall meet the requirements shown in table II. The performance test results shall meet the

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requirements shown in table III. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.3(v)).

7. NOTES

7.1 Sources of documents

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

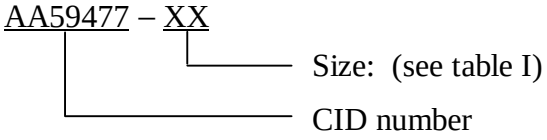
7.1.3 ISO standards. Copies of ISO standards may be obtained from the International Organization for Standardization, 1, Rue de Varembé, Case Postale 56, CH-1211 Genéve 20, Switzerland.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.6 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



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7.3 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Size of machine required, if different (see 2)
c. Optional cooling equipment, as specified (see 3.2.3)
d. Grinding wheel infeed, if different (see 3.2.4)
e. Regulating wheel infeed, if different (see 3.2.5)
f. Range of dressing depth, if different (see 3.2.6)
g. Dresser, if different (see 3.2.6)
h. Size and quantity of workrest blades, specify (see 3.2.7)
i. Regulating wheel (12 inches and less) tilt and/or swivel, if different (see 3.2.9)
j. Regulating wheel (12 inches and greater) tilt and/or swivel, if different (see 3.2.9)
k. Electrical requirements, if different (see 3.2.10)
l. Dripproof enclosures, if different (see 3.2.11)
m. Optional equipment, if required (see 3.2.12)
n. Machine dimensions (length, width, and height) and weight, if different (see 3.4)
o. Measurement system, specify (see 3.6)
p. Dual graduation, if required (see 3.6)
q. Nameplate, if different (see 3.7)
r. Lubrication plate, if different (see 3.8)
s. Testing, if different (see 5.2)
t. Environmental conditions for test, if different (see 5.2.1)
u. Acceptance test, if different (see 5.3)
v. Packaging requirements (see 6.1)

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS
(Project 3415-0208)

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